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Predation increases species turnover and structural stability in ecological communities

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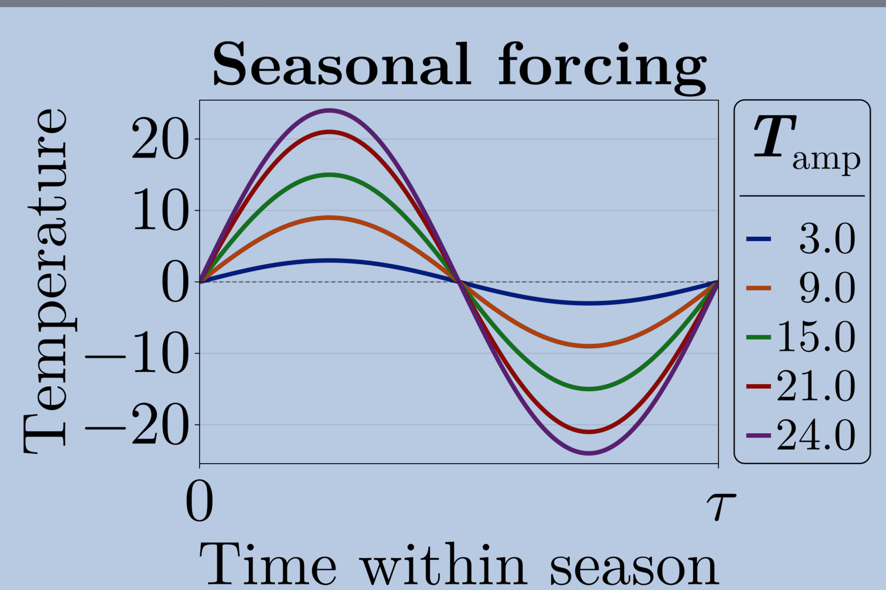
Background

A mechanistic understanding of species coexistence is a central pursuit in ecology. Prior eco-evolutionary models show that seasonal temperature fluctuation can facilitate coexistence through the *storage effect* – species with differing traits can coexist if their gains in good times exceed their losses in bad times. However, the role of predation in shaping coexistence and diversity is less well understood. We compare two models, with and without predation, to explore its effects on emergent community structure and stability.

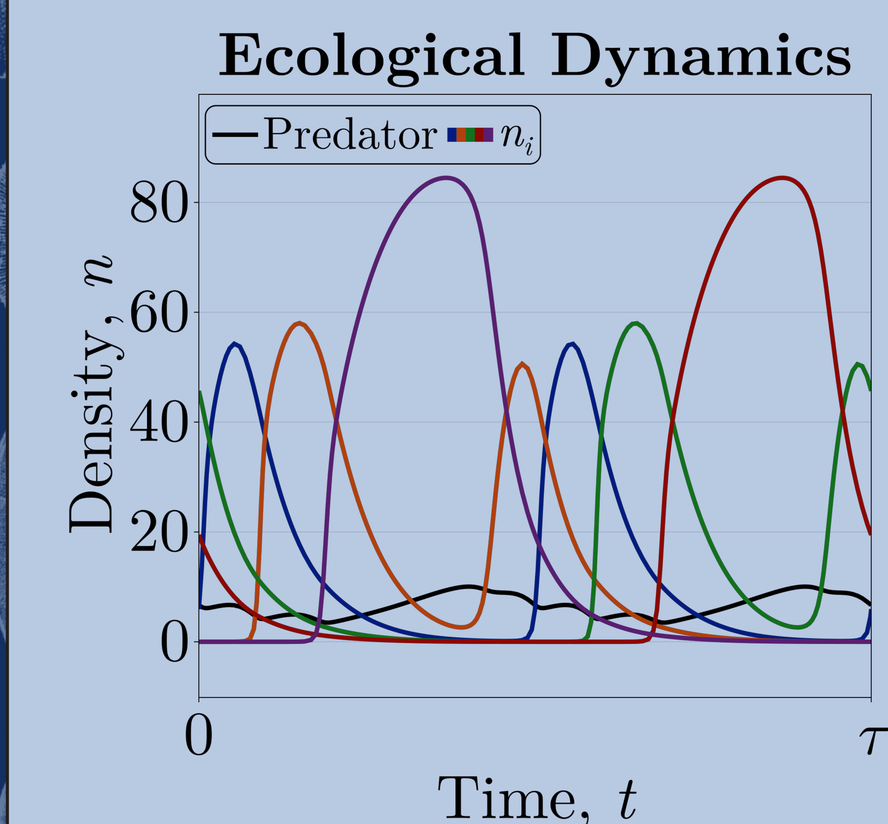
Methods



How temp. changes

Our model uses seasonal temperature fluctuation (T_{amp}) to study the evolution of thermal optima and its impact on species coexistence.

Fig. 1. How temperature is modeled over one season (τ), for several amplitudes (T_{amp}).



Finding a stable seasonal cycle

First we simulate the density of our community over time. After a long simulation we find a periodic solution to the ecological dynamics, which repeats with the seasonal temperature.

Fig. 2. Species densities once the community settles into a repeating seasonal pattern (a limit cycle), shown over one full season (τ).

Survival of the fittest

Next we allow thermal optima to evolve. These seed communities eventually reach an equilibrium where no other species can invade. We call these *Evolutionarily Stable Communities* (ESCs).

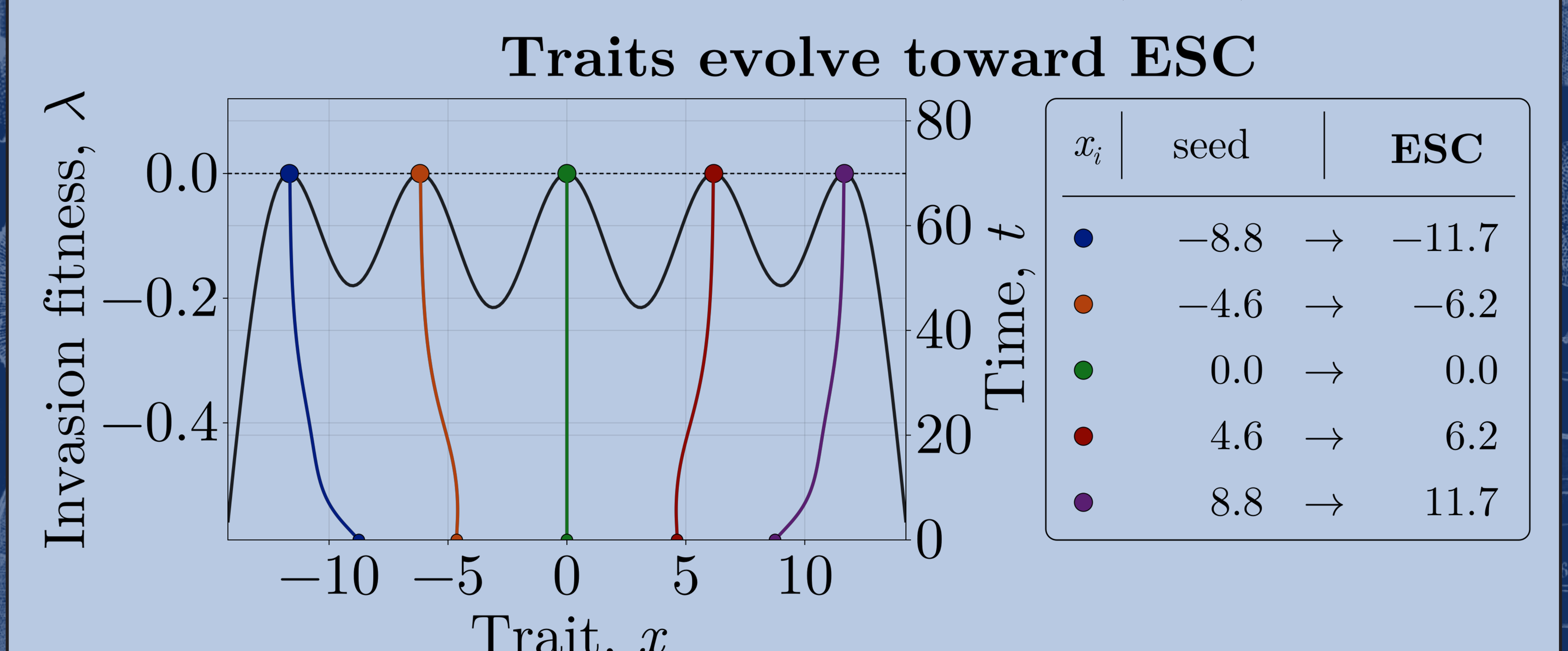


Fig. 3. Thermal optima (x) evolving from a scattered start; each climbs the fitness landscape (λ) to a peak, where no new species can invade (an ESC).

Results

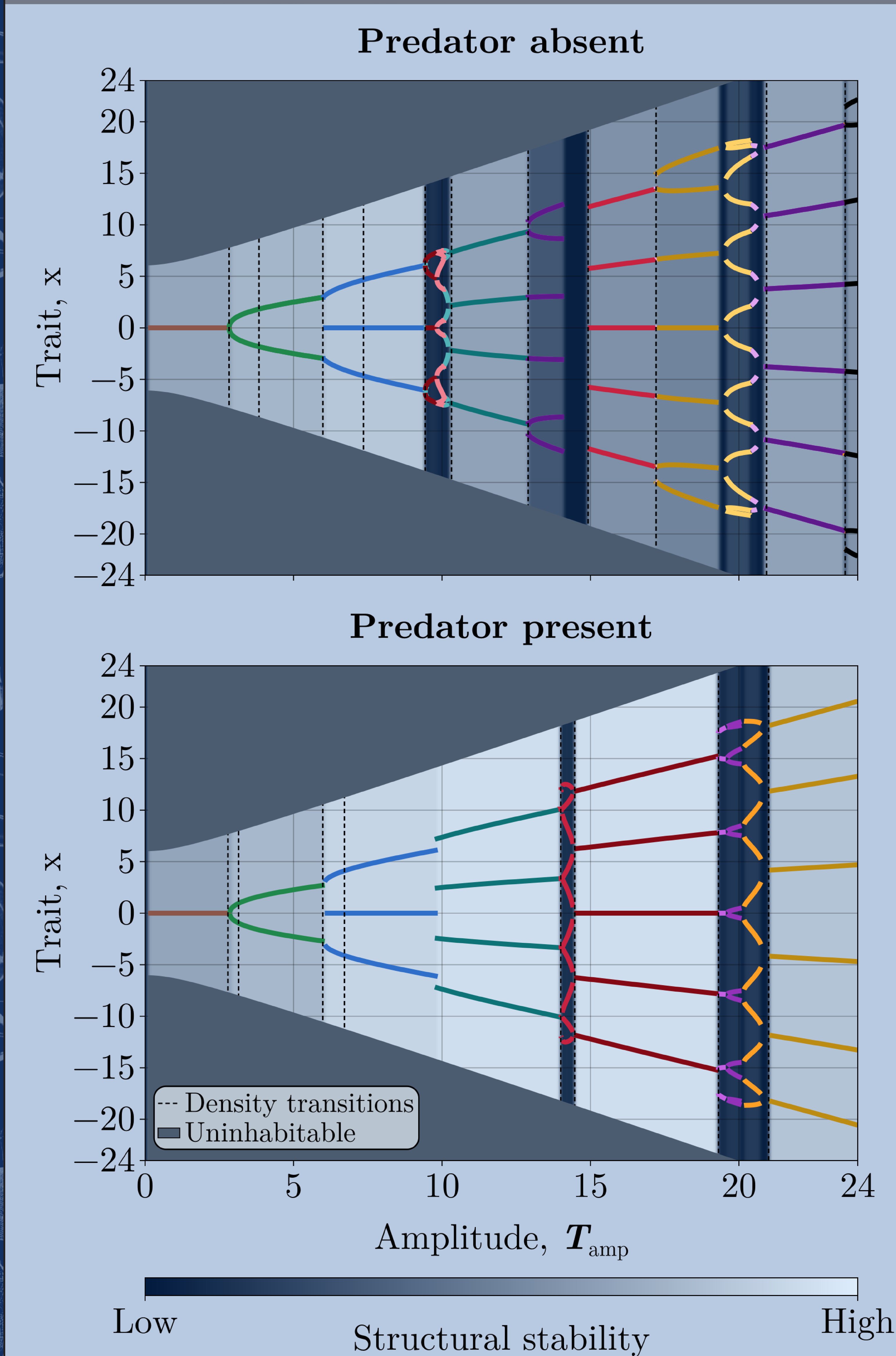


Fig. 4. Community bifurcation diagrams: evolutionarily stable community (ESC) trait values across seasonal amplitude (T_{amp}). Top: predator absent. Bottom: predator present. Shading highlights structural stability: lighter represents a relatively wider range of environments over which the community persists. Vertical dashed lines mark qualitative transitions (by eye) in species densities; the small intervals between these dashed lines represent regions where species densities are quickly changing.

Diversity

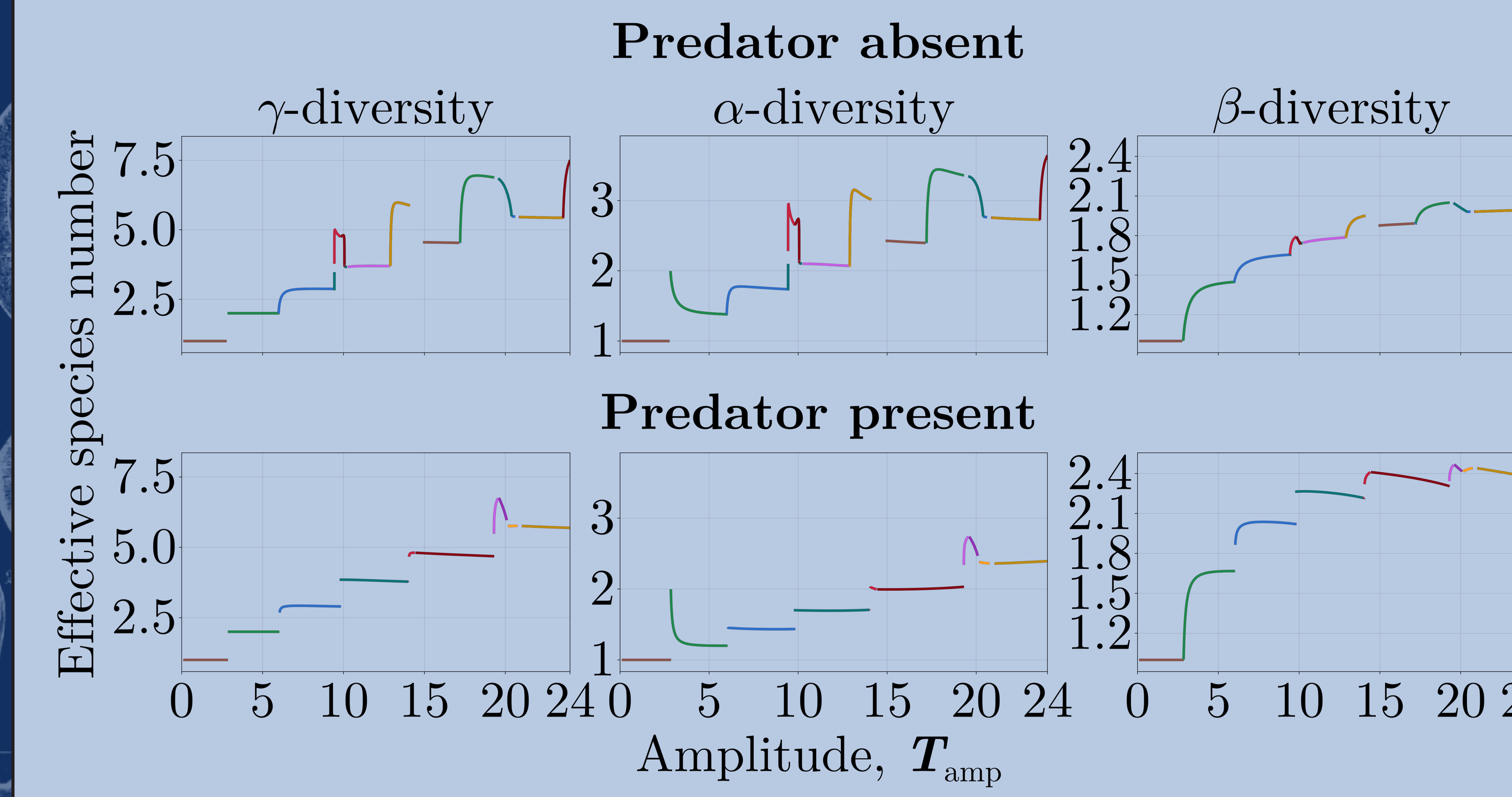


Fig. 5. Species diversity across seasonal amplitude (T_{amp}). Top: predator absent. Bottom: predator present. α-diversity is average instantaneous species diversity, γ-diversity is time-integrated species diversity, and β-diversity is species turnover. Predation decreases α alongside a relatively smaller decrease in γ; the shift in their ratio drives higher average turnover, β.

Discussion

- **Predation increases structural stability, widening the range of environments under which communities can stably coexist.**
- **Introducing a predator increases species turnover, β.**
- **Predation supports more species richness at lower temperature amplitudes ecologically, but these communities fail to persist once thermal optima are allowed to evolve.**

Our study implies that predation can foster communities that are more robust to varying amplitudes of temperature fluctuations, raising an exciting empirical question: does predation allow communities to remain qualitatively unchanged as temperature fluctuations intensify?

Acknowledgments

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Scan for digital twin, references, and further reading

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